

Automated Retinal Blood Vessel Segmentation

Using UNet++ Architecture

Thesis Defense Presentation

IIT (BHU)

Introduction & Motivation

- Clinical Importance: Retinal blood vessels are key indicators for diagnosing cardiovascular
 - and ophthalmological diseases (e.g., Diabetic Retinopathy).
- The Challenge: Manual segmentation by clinicians is time-consuming, subjective, and
 - highly prone to human error.
- The Complexity: Retinal images feature fine, low-contrast micro-capillaries that are
 - incredibly difficult to extract precisely using traditional image processing.
- Objective: Develop an automated, highly precise deep learning model capable of
 - extracting the full vascular network from RGB fundus images.

The Dataset & Preprocessing

- Source: Annotated clinical dataset of High-Resolution RGB Fundus images.
- Volume: 137 images split into training, validation, and testing sets.
- Ground Truth: Precise binary masks manually annotated by clinical experts.
- Preprocessing Pipeline:
 - Resize to unified dimensions of 256 x 256 for network compatibility.
 - Min-Max Normalization to scale pixel values into the [0, 1] range.

Methodology: UNet++ & Loss Function

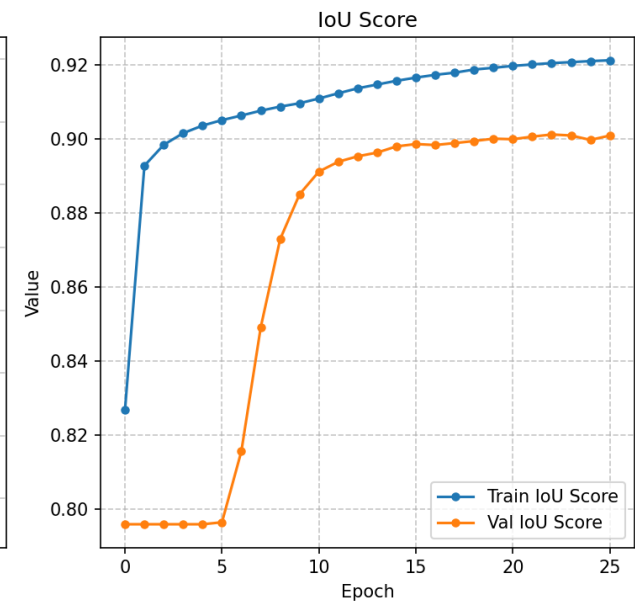
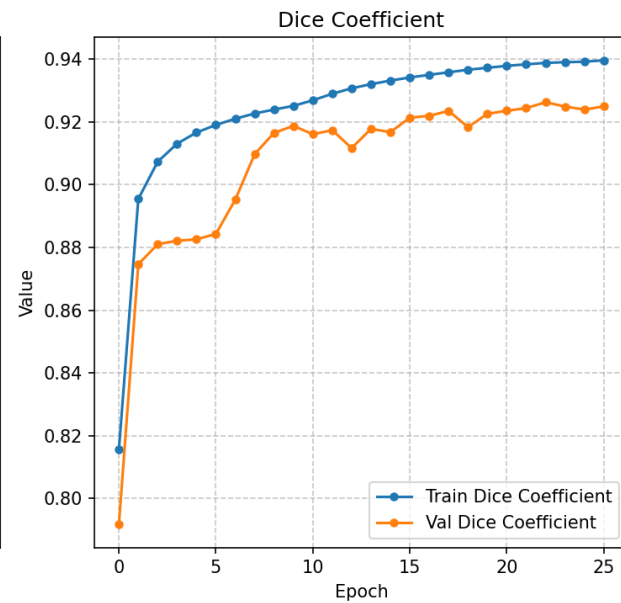
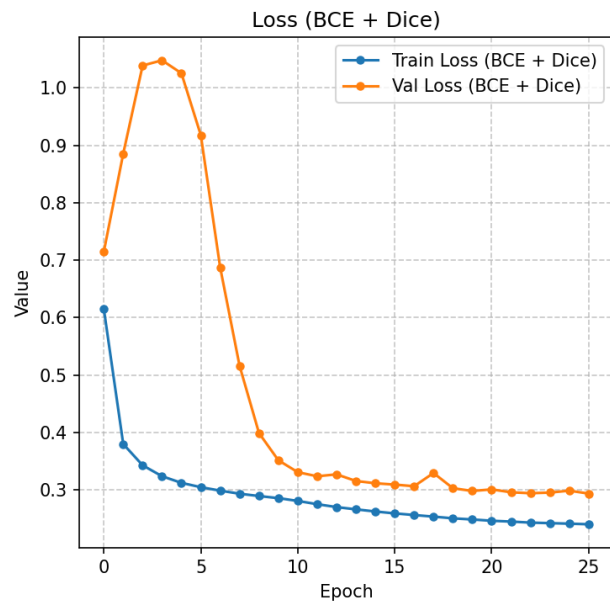
- Architecture: UNet++
 - A deeply supervised encoder-decoder network.
 - Features nested, dense skip pathways to bridge the semantic gap.
 - Vastly improved capture of both thick primary vessels and ultra-thin capillaries.
- Loss Function Formulation:
 - Loss = Binary Cross-Entropy (BCE) + Dice Loss
 - BCE: Ensures stable pixel-wise probability distributions.
 - Dice Loss: Solves severe class imbalance by directly maximizing Intersection over Union.

Experimental Setup

- Framework: Built in TensorFlow 2.x and Keras.
- Optimizer: Adam Optimizer for robust gradient updates.
- Learning Rate Scheduling:
 - ReduceLROnPlateau callback implemented to decay LR automatically when metrics stagnate.
- Metrics Tracked:
 - Dice Coefficient
 - Intersection over Union (IoU)
 - Accuracy, Precision, Recall

Results: Training Metrics

- The model reached stable convergence over 26 epochs.
- Best Validation Dice Coefficient: ~0.926
- Best Validation IoU Score: ~0.901

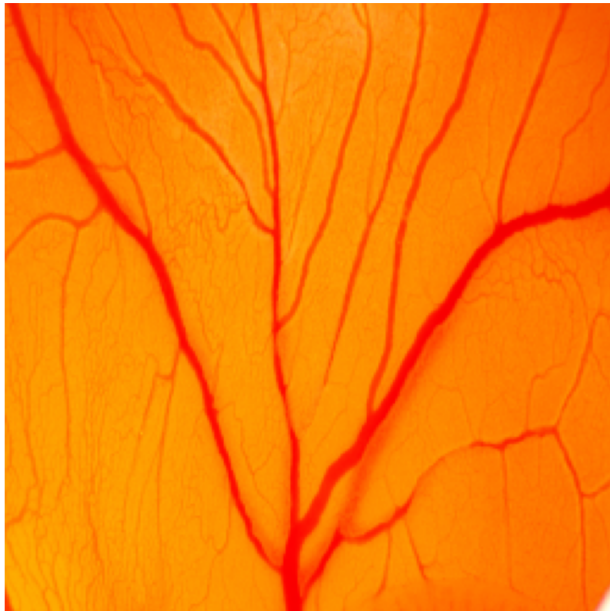


Results: Visual Inference Output

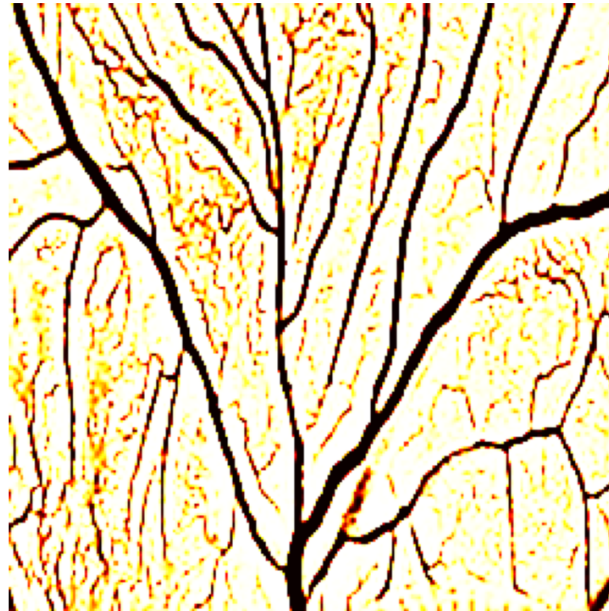
- Prediction on an unseen test sample (Image 501):
- Red overlay indicates predicted vessel boundaries matching the underlying image.

UNet++ Prediction — Image 501

RGB Input



Prediction (heatmap)



Overlay (red = predicted boundary)



Conclusion & Future Work

- Conclusion:

- UNet++ coupled with a hybrid BCE+Dice loss achieves exceptional precision in
- retinal vessel segmentation.
- Successfully captures both macro and micro vascular structures.

- Future Work:

- Integration of Spatial Attention Modules (Attention UNet++).
- Multi-class classification to distinguish between Arteries and Veins.
- Cross-dataset generalization testing (e.g., DRIVE or STARE datasets).